UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Parallel and multiprocessor architecture

a)
Describe ONE (1) major difference between a tightly coupled multiprocessor system and a loosely coupled multiprocessor system.
b)
State the THREE (3) types of cluster architecture.

Source: CSC429 (12).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Data representation and number systems

Convert 100.5 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: 100.510 = 0x42C90000 (sign=0, exponent=10000101, fraction=10010010000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A3 (5 marks)

Topic: Computer generations and history

List the name of the inventor of the following devices:
a) Mechanical Calculating Machine
b) Difference Machine
c) Analytical Machine
d) IAS Machine
e) Integrated circuit

Source: CSC429 (14).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

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QUESTION A5 (10 marks)

Topic: Input/output and interrupts

a) Why must the CPU be faster than all the other inpuuoutput devices?
b) How do the I/O controllers facilitate communication between the CPU and I/O devices?
c) Discuss TWO (2) types of I/O modules.
d) When would an I/O controller of a specific device be installed?

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.
a) Convert 159910 to hexadecimal.
b) Convert FD916 to decimal.

a) Convert 159910 to hexadecimal.

Answer: 159910 = 63F16

- *1 mark for repeated division by 16.*
- *1 mark for correct remainders.*
- *1 mark for reading remainders in reverse order.*
- *1 mark for the correct hexadecimal digits.*
- *1 mark for the base subscript.*

b) Convert FD916 to decimal.

Answer: FD916 = 405710

- *1 mark for place values 16^2, 16^1 and 16^0.*
- *1 mark for correct digit values.*
- *1 mark for correct products.*
- *1 mark for correct addition.*
- *1 mark for the final decimal value.*

QUESTION A7 (10 marks)

Topic: Input/output and interrupts

a) Discuss THREE (3) types of interfaces to extemal devices.
b) Classify the following input/output devices to their respective 1/0 module:
i)
Network interface card
ii)
Scanner
Touch Screen
iv)
Register

Source: CSC429 (7).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

System interconnection is a mechanism that provides communication among computer components.
a) What is the size of a bus if the word size is 64 bit?
b) List the TWO (2) types of bus.
c) Illustrate the location of a bus between CPU and memory.
d) Differentiate between system interconnection and CPU interconnection.
e) Discuss ONE (1) drawback each for the different types of bus timing.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Memory hierarchy and cache

a)
Discuss the main function of cache memory.
b)
How can the data stored in the cache and main memory be synchronized.
c)
Contrast between SRAM and DRAM based on the following characteristics:
Type of storage
Availability of charges leak
price
Example of location of memory usage

Source: CSC429 (15).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Computer architecture and CPU organisation

a) Draw a complete machine cycle by giving the exact location of each phase.
b) Briefly, describe each phase in the machine cycle.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		
MOV (p,q)		
ADD (p.q.r)		
STORp		
ADD (p,q)		
SUB (p,q,r)		
ADDp		
SUB (p,q)		
MUL (p.q.r)		
SUBp		
MUL (p,q)		
DIV(p,q,r)		
MULp		
DIV(p,q)		
DIVp		

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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